

CS472 Software Testing and Continuous Integration Lab Report

Ethan Guillem

Group 2 Individual Report

Introduction

This report documents my individual work across test coverage, test driven development, and continuous integration. Each phase followed the team contribution workflow: synchronize the fork, define the work in an issue, implement on a focused branch, preserve test or CI evidence, open a pull request, and complete an evidence based peer review.

My Phase 1, Phase 2, and Phase 3 substantive pull requests received the required teammate approval and were merged into the team repository. The report below includes the implementation, test and coverage results, RED-GREEN-REFACTOR evidence, CI failure diagnosis and recovery, peer reviews, reviewer feedback, and final merge evidence.

Repository and review assignments

- Fork repository: [Ethan Guillem fork](#)
- Team upstream repository: [Group 2 upstream](#)
- My assigned reviewer for substantive pull requests: Jonathan Johnson, GitHub johns216
- My assigned review author: Astrid Jimenez, GitHub jimenaz29

Phase 1 Test Coverage

Setup and environment

I synchronized my fork with the team upstream, verified the test_coverage_lab folder, installed the required dependencies, and ran pytest with coverage. The team setup contribution was merged before my individual work began.

- Setup issue: [Phase 1 setup issue 43](#)
- Setup pull request: [Phase 1 setup PR 44](#)
- Verification command: `pytest --cov=models --cov-report=term-missing`
- The starter files were stored under test_coverage_lab with no nested Git repository.

```
ethanguillem@Ethans-MacBook-Pro Group-2 % cd /Users/ethanguillem/cs472/Group-2/t
tests requirements.txt && .venv/bin/python -m pytest --cov=models --cov-report=term-missing
/Users/ethanguillem/cs472/Group-2/test_coverage_lab
models requirements.txt tests
===== test session starts =====
platform darwin -- Python 3.14.6, pytest-8.4.2, pluggy-1.6.0 -- /Users/ethanguillem/cs472/Group-2/test_coverage_lab/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/ethanguillem/cs472/Group-2/test_coverage_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 7 items

tests/test_account.py::test_account_role_assignment PASSED [ 14%]
tests/test_account.py::test_invalid_role_assignment PASSED [ 28%]
tests/test_account.py::test_to_dict PASSED [ 42%]
tests/test_account.py::test_required_fields PASSED [ 57%]
tests/test_account.py::test_password_hashing PASSED [ 71%]
tests/test_account.py::test_account_deactivation_and_reactivation PASSED [ 85%]
tests/test_account.py::test_valid_withdrawal PASSED [100%]

===== tests coverage =====
coverage: platform darwin, python 3.14.6-final-0

Name                Stmts  Miss  Cover   Missing
-----
models/__init__.py      7      0  100%
models/account.py      58     14    76%   30, 47-49, 60-62, 66-68, 73, 75, 102-103
tests/__init__.py       0      0  100%
tests/test_account.py   69      0  100%
-----
TOTAL                  134     14    90%

===== 7 passed in 0.40s =====
ethanguillem@Ethans-MacBook-Pro test_coverage_lab %
```

Local setup verification: test_coverage_lab on my Mac; 7 tests passed, 90% total coverage. This later run uses the Phase 2 checkout and does not include my positive-deposit test. PR 78 below records that test and its original before/after coverage.

Issue and assigned test

I claimed Pool Item 4, testing a positive account deposit. The acceptance criteria required a known starting balance, a positive deposit, an exact resulting balance assertion, successful local tests, and evidence that the previously uncovered deposit path was exercised.

- Issue and acceptance criteria: [Phase 1 issue 71](#)
- Substantive contribution: [Phase 1 PR 78](#)
- Test source: [positive deposit test file](#)

Test implementation

The test creates an Account with a balance of 100.00, deposits 50.00, and asserts that the exact resulting balance is 150.00. This exercises the successful path in Account.deposit and provides a meaningful behavioral assertion.

```
# Test: Positive Account Deposit
```

```
# Author: Ethan Guillem
```

```
# Description: Ensure a positive deposit increases the balance correctly.
```

```
def test_positive_deposit():  
    account = Account(  
        name="Ethan Guillem",  
        email="ethan@example.com",  
        balance=100.00,  
    )  
    account.deposit(50.00)  
    assert account.balance == 150.00
```

Test and coverage evidence

Targeted test

```
tests/test_account.py::test_positive_deposit PASSED
```

```
1 passed
```

Complete suite

```
6 passed
```

Coverage before the new test

```
models/account.py 67%
```

```
TOTAL 82%
```

Coverage after the new test

```
models/account.py 71%
```

```
tests/test_account.py 100%
```

```
TOTAL 86%
```

The contribution increased models/account.py coverage from 67 percent to 71 percent and total coverage from 82 percent to 86 percent while keeping the full Phase 1 suite green.

GitHub Actions evidence: [Phase 1 successful CI run](#)

Technical peer review

I reviewed Astrid Jimenez's insufficient funds withdrawal test. I verified that the test called withdraw with an amount greater than the balance, expected DataValidationError, preserved the balance, and remained focused on the linked issue.

- Reviewed pull request: [Astrid Phase 1 PR 65](#)
- Initial evidence based changes requested: [Phase 1 changes requested review](#)
- Final approval after corrections: [Phase 1 approval review](#)
- The author added the required docstring, synchronized with main, corrected the file path in the PR description, and updated the test evidence.
- The reviewed PR was approved and merged.

Feedback and final status

Jonathan Johnson approved the Phase 1 pull request after confirming that the positive-deposit test exercised the intended behavior, used a correct assertion, increased coverage, and satisfied issue 71. He requested no changes, so no additional revision was necessary. [Phase 1 approval review](#). [Phase 1 merge commit](#).

Phase 2 Test Driven Development

Setup and environment

I synchronized the fork after the TDD setup pull request was merged, verified the `tdd_lab` project, and studied the worked create counter example before beginning my feature.

- Setup issue: [Phase 2 setup issue 46](#)
- Setup pull request: [Phase 2 setup PR 47](#)
- Verification command: `pytest --cov=src`

```
ethanguille@Ethans-MacBook-Pro Group-2 % cd /Users/ethanguille/cs472/Group-2/t
ments.txt && .venv/bin/python -m pytest --cov=src --cov-report=term-missing
/Users/ethanguille/cs472/Group-2/tdd_lab
requirements.txt      src      tests
===== test session starts =====
platform darwin -- Python 3.14.6, pytest-8.4.2, pluggy-1.6.0 -- /Users/ethanguille/cs472/Group-2/tdd_lab/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/ethanguille/cs472/Group-2/tdd_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 6 items

tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 16%]
tests/test_counter.py::TestCounterEndpoints::test_delete_counter PASSED [ 33%]
tests/test_counter.py::TestCounterEndpoints::test_reject_invalid_counter_name PASSED [ 50%]
tests/test_counter.py::TestCounterEndpoints::test_list_counters PASSED [ 66%]
tests/test_counter.py::TestCounterEndpoints::test_retrieve_existing_counter PASSED [ 83%]
tests/test_counter.py::TestCounterEndpoints::test_reset_all_counters PASSED [100%]

===== tests coverage =====
coverage: platform darwin, python 3.14.6-final-0

Name                Stmts  Miss  Cover   Missing
-----
src/_init_.py         3      0  100%
src/counter.py       32      1   97%    33
src/status.py         7      0  100%
TOTAL                 42      1   98%

===== 6 passed in 0.11s =====
```

Local setup verification: `tdd_lab` on my Mac; 6 tests passed, 98% total coverage. The screenshot includes `test_reject_invalid_counter_name` passing.

Issue and feature

I claimed Feature Pool Item 1, rejecting non alphanumeric counter names submitted to `POST /counters/<name>`. The endpoint now returns HTTP 400 and a clear JSON error while continuing to accept valid alphanumeric names.

- Issue and acceptance criteria: [Phase 2 issue 79](#)
- TDD contribution: [Phase 2 PR 80](#)

RED

I wrote the invalid name test before changing the implementation. The existing route incorrectly accepted invalid-name and returned HTTP 201 instead of HTTP 400.

RED commit: [057eb38 test reject invalid counter names](#)

```
test_reject_invalid_counter_name FAILED
```

```
E      assert 201 == 400
```

```
1 failed
```

This failure proved that the endpoint lacked name validation and that the test was exercising the intended missing behavior.

GREEN

The minimum implementation added HTTP_400_BAD_REQUEST, checked the name before creating the counter, and returned a clear JSON error.

GREEN commit: [9e9e521 reject invalid names](#)

```
def test_reject_invalid_counter_name(self, client):
    invalid_name = "invalid-name"
    result = client.post(f"/counters/{invalid_name}")
    assert result.status_code == HTTPStatus.BAD_REQUEST
    assert result.get_json() == {
        "error": f"Invalid counter name: {invalid_name}"
    }
    if not name.isalnum():
        return jsonify(
            {"error": f"Invalid counter name: {name}"},
            status.HTTP_400_BAD_REQUEST
```

The new test and the existing tests passed after this minimum implementation. Test source: [invalid counter name test file](#).

REFACTOR

I extracted the validation rule into `is_valid_counter_name` so the route remains readable and the rule can be reused. The API behavior did not change.

REFACTOR commit: [cb7152f extract validation helper](#)

```
def is_valid_counter_name(name):
    return name.isalnum()
```

Final post synchronization verification

6 tests passed

src/counter.py	97%
----------------	-----

TOTAL	98%
-------	-----

Current CI evidence: [Phase 2 successful matrix run](#)

Technical peer review

I reviewed Astrid Jimenez's increment counter feature. I examined the linked issue, the RED evidence showing HTTP 405 before the route existed, the GREEN implementation returning the incremented value, the no refactor justification, coverage, and the green CI result.

- Reviewed pull request: [Astrid Phase 2 PR 67](#)
- Approval: [Phase 2 approval review](#)

The feature test matched the issue, the RED and GREEN sequence was supported by evidence, the final implementation remained focused, and the existing suite passed.

Feedback and final status

Jonathan Johnson approved the Phase 2 pull request after confirming that the test matched issue 79, demonstrated a genuine RED failure before the implementation, passed after the GREEN change, improved coverage, and satisfied the acceptance criteria. He requested no changes, so no additional revision was necessary. [Phase 2 approval review](#). [Phase 2 merge commit](#).

Phase 3 Continuous Integration

Repository setup

I synchronized my fork, verified ci_lab locally, and confirmed that the starter workflow was stored at .github/workflows/ci.yml with ci_lab as its working directory.

- Fork repository: [Ethan Guillem fork](#)
- Setup issue: [Phase 3 setup issue 51](#)
- Setup pull request: [Phase 3 setup PR 52](#)
- Local baseline: 22 tests passed with 95 percent total coverage.

```
pytest --cov=src --cov-report=term-missing
22 passed
src/counter.py      94%
TOTAL               95%
```

```
ethanguillem@Ethans-MacBook-Pro Group-2 % cd /Users/ethanguillem/cs472/Group-2/c
ents.txt && .venv/bin/python -m pytest --cov=src --cov-report=term-missing
/Users/ethanguillem/cs472/Group-2/ci_lab
requirements.txt      src      tests
===== test session starts =====
platform darwin -- Python 3.14.6, pytest-8.4.2, pluggy-1.6.0 -- /Users/ethanguillem/cs472/Group-2/ci_lab/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/ethanguillem/cs472/Group-2/ci_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 22 items

tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 4%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_duplicate_counter PASSED [ 9%]
tests/test_counter.py::TestCounterEndpoints::test_retrieve_existing_counter PASSED [ 13%]
tests/test_counter.py::TestCounterEndpoints::test_return_404_for_non_existent_counter PASSED [ 18%]
tests/test_counter.py::TestCounterEndpoints::test_increment_counter PASSED [ 22%]
tests/test_counter.py::TestCounterEndpoints::test_prevent Updating non-existent counter PASSED [ 27%]
tests/test_counter.py::TestCounterEndpoints::test_delete_counter PASSED [ 31%]
tests/test_counter.py::TestCounterEndpoints::test_prevent deleting non-existent counter PASSED [ 36%]
tests/test_counter.py::TestCounterEndpoints::test_reset_all_counters PASSED [ 40%]
tests/test_counter.py::TestCounterEndpoints::test_list_all_counters PASSED [ 45%]
tests/test_counter.py::TestCounterEndpoints::test_handle_invalid http methods PASSED [ 50%]
tests/test_counter.py::TestCounterEndpoints::test_get_total counters PASSED [ 54%]
tests/test_counter.py::TestCounterEndpoints::test_top_n counters PASSED [ 59%]
tests/test_counter.py::TestCounterEndpoints::test_bottom_n counters PASSED [ 63%]
tests/test_counter.py::TestCounterEndpoints::test_set counter to value PASSED [ 68%]
tests/test_counter.py::TestCounterEndpoints::test_prevent negative counter values PASSED [ 72%]
tests/test_counter.py::TestCounterEndpoints::test_reset single counter PASSED [ 77%]
tests/test_counter.py::TestCounterEndpoints::test_prevent resetting non-existent counter PASSED [ 81%]
tests/test_counter.py::TestCounterEndpoints::test_get total number of counters PASSED [ 86%]
tests/test_counter.py::TestCounterEndpoints::test_counters greater than threshold PASSED [ 90%]
tests/test_counter.py::TestCounterEndpoints::test_counters less than threshold PASSED [ 95%]
tests/test_counter.py::TestCounterEndpoints::test_validate counter name PASSED [100%]

===== tests coverage =====
_____ coverage: platform darwin, python 3.14.6-final-0 _____

Name                Stmts   Miss  Cover   Missing
-----
src/_init_.py         3      0   100%
src/counter.py       87      5    94%   71, 85, 96, 99-100
src/status.py         7      0   100%
TOTAL                97      5    95%

===== 22 passed in 0.14s =====
```

```
ethanguillem@Ethans-MacBook-Pro ci_lab %
```

Local setup verification: ci_lab on my Mac; all 22 starter tests passed, with 94% coverage for src/counter.py and 95% total coverage.

Combined CI contribution

I claimed Enhancement 2, dependency caching. The workflow caches pip downloads using actions/cache, keys the cache by runner operating system and the ci_lab requirements file hash, and provides an operating system specific restore key.

- Issue and acceptance criteria: [Phase 3 issue 81](#)
- Combined CI contribution: [Phase 3 PR 82](#)
- Workflow source: [dependency caching workflow](#)

```
- name: Cache Python Dependencies
  uses: actions/cache@v4
  with:
    path: ~/.cache/pip
    key: runner OS plus requirements file hash
    restore-keys: operating system pip prefix
```

Controlled failure and diagnosis

The controlled failure intentionally referenced actions/cache@v99. GitHub Actions failed during job setup because no v99 release exists, before checkout, dependency installation, or tests could run.

- Controlled failure commit: [a8f6b3b](#)
- Failing CI log: [controlled failure run](#)

```
Unable to resolve action actions/cache@v99
Unable to find version v99
```

Root cause: the action resolver could not locate the intentionally invalid release tag. The cache path, key design, Python dependencies, and test suite were not the cause.

Fix and successful reruns

The focused fix changed only actions/cache@v99 to the supported actions/cache@v4 reference. The first successful run missed because the cache did not exist yet, then saved it. A rerun restored the exact cache key.

- Fix commit: [9d23fa5](#)
- Successful recovery run: [cache save run](#)
- Successful subsequent run: [cache hit run](#)

```
Cache hit for: Linux-pip-24b8f8c...
Cache restored successfully
22 passed
TOTAL 95%
```

Technical peer review

I reviewed Astrid Jimenez's CI visibility and documentation enhancement. I verified the README diff, badge target, documented pytest coverage gate and Flake8 checks, controlled failure logs, root cause, corrective commit, and successful matrix runs on Python 3.9, 3.10, and 3.11.

- Reviewed pull request: [Astrid Phase 3 PR 88](#)
- Approval: [Phase 3 approval review](#)
- The reviewed pull request was approved and merged.

Feedback and final status

Jonathan Johnson approved the Phase 3 pull request after confirming that dependency caching was implemented correctly, the controlled failure was clearly demonstrated, the root-cause analysis matched the CI logs, the fix restored a successful run, and the acceptance criteria were satisfied. He requested no changes, so no additional revision was necessary. [Phase 3 approval review](#). [Phase 3 merge commit](#).

Reflection

This lab showed me that automated testing is most useful when it is tied to a repeatable engineering workflow. Coverage helped identify code that was not exercised, but the positive deposit test also needed a specific assertion to prove the balance changed by the correct amount. A high percentage alone would not establish that behavior.

The TDD phase made the RED step concrete. Capturing the initial HTTP 201 result proved that invalid names were actually accepted before implementation. The GREEN change stayed small, and the REFACTOR step improved reuse without altering the response. Keeping separate commits made the development sequence easy to review.

The CI phase demonstrated how quality gates and workflow configuration turn local practices into team rules. The controlled failure was useful because the Actions log isolated the failure to an invalid action version. After correcting the version, the cache save and later cache hit showed that the enhancement worked rather than merely passing YAML validation.

Peer review also depended on evidence rather than general approval. I used test behavior, coverage output, workflow diffs, and CI logs to evaluate teammate contributions. Merge coordination mattered because several branches modified the same test and workflow files. Synchronizing after upstream merges prevented one contribution from overwriting another.

Final submission checklist

- Completed: Local setup screenshots are included in all three phase sections.
- Completed: Jonathan Johnson approved PR 78, PR 80, and PR 82; the feedback links are included in each phase.
- Completed: Final merge commit links are included for all three individual contributions.
- Completed: PR 80 was approved and merged.
- Confirm the fork is public and every evidence label opens the intended GitHub page.
- Export this document to one PDF, commit the PDF to the team repository, and upload the same PDF to Canvas.